History-derived proxy for background discrepancy

| | H1 | H2 | H3 | H4 | H5 | H6 | H7 | H8 | H9 | H10 | H11 | H12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Low n = 79.0k–82.6k | +2.175 | +2.355 | +3.074 | +2.352 | +1.287 | +1.918 | +2.001 | +2.575 | +2.859 | +3.643 | +2.639 | +2.232 |
| Medium n = 99.4k–102.1k | +1.405 | +1.549 | +2.075 | +1.602 | +0.752 | +1.269 | +1.346 | +1.739 | +1.945 | +2.536 | +1.784 | +1.465 |
| High n = 126.5k–130.1k | +0.639 | +0.725 | +1.051 | +0.917 | +0.381 | +0.653 | +0.645 | +0.844 | +0.949 | +1.193 | +0.839 | +0.606 |

**(a) Chengdu · inflow**

| | H1 | H2 | H3 | H4 | H5 | H6 | H7 | H8 | H9 | H10 | H11 | H12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Low n = 85.8k–88.8k | +1.594 | +1.316 | +0.826 | +1.089 | +0.244 | +0.481 | +0.853 | +1.419 | +0.732 | +2.018 | +1.127 | +0.483 |
| Medium n = 111.3k–115.2k | +1.063 | +0.911 | +0.543 | +0.617 | +0.073 | +0.184 | +0.605 | +0.891 | +0.486 | +1.430 | +0.739 | +0.256 |
| High n = 114.9k–119.5k | +0.450 | +0.386 | +0.162 | +0.233 | -0.005 | +0.069 | +0.276 | +0.435 | +0.222 | +0.725 | +0.312 | +0.115 |

**(b) Chengdu · outflow**

| | H1 | H2 | H3 | H4 | H5 | H6 | H7 | H8 | H9 | H10 | H11 | H12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Low n = 91.4k–95.1k | +14.749 | +14.970 | +14.310 | +14.557 | +14.673 | +13.620 | +14.213 | +13.691 | +13.984 | +14.263 | +14.574 | +14.060 |
| Medium n = 112.5k–115.9k | +10.839 | +11.060 | +10.733 | +10.488 | +10.588 | +9.982 | +10.529 | +10.337 | +10.439 | +10.113 | +10.912 | +10.191 |
| High n = 129.0k–131.0k | +7.719 | +7.978 | +7.883 | +7.787 | +7.956 | +7.434 | +7.869 | +7.903 | +8.132 | +7.395 | +8.098 | +7.605 |

**(c) Shanghai · inflow**

| | H1 | H2 | H3 | H4 | H5 | H6 | H7 | H8 | H9 | H10 | H11 | H12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Low n = 90.6k–93.4k | +13.366 | +13.568 | +12.795 | +13.505 | +13.686 | +12.536 | +13.042 | +12.368 | +12.431 | +13.159 | +13.325 | +12.223 |
| Medium n = 110.7k–114.2k | +9.663 | +9.842 | +9.510 | +10.018 | +9.874 | +9.245 | +9.701 | +9.206 | +9.226 | +9.612 | +9.756 | +8.938 |
| High n = 131.0k–134.2k | +7.105 | +7.053 | +6.928 | +7.228 | +7.206 | +6.801 | +7.267 | +6.703 | +6.913 | +6.997 | +7.333 | +6.365 |

**(d) Shanghai · outflow**

